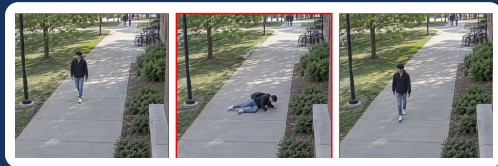




University of Dhaka
Department of Computer Science and Engineering

ARGUS

An Adaptive, Resource-Aware,
Grounded Understanding System for
Real-Time Explainable Video Anomaly
Detection



normal → **anomaly** → normal

Md. Shahriar Kabir

Exam Roll: CURZON-2323816

Reg. No: 2021-711-196

Tawyabul Islam Tamim

Exam Roll: CURZON-2323826

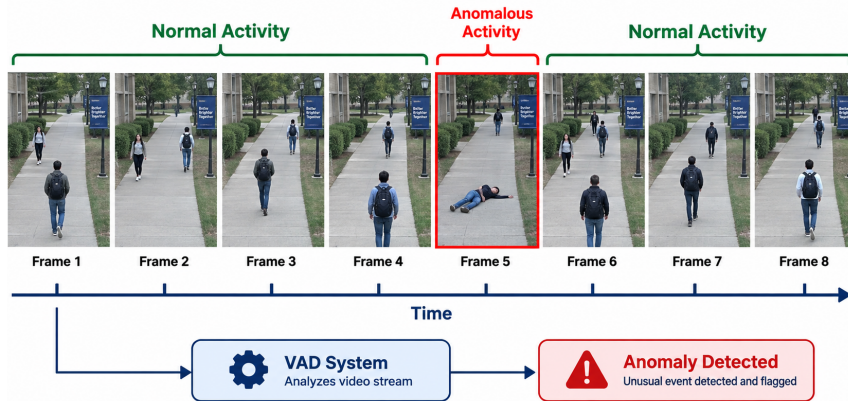
Reg. No: 2021-411-180

Supervisor: Md. Ahasanul Alam

Contents

- 1 Introduction** the problem, why it matters, and our objectives
- 2 Related Works** what exists, and what is still missing
- 3 Methodology** how ARGUS decides, answers and explains
- 4 Results** what the simulation already shows
- 5 Summary** what is borrowed, and what is new

What is video anomaly detection?



Watch **normal activity**, and flag the **unusual** moment – **while it happens**.

Problem definition

The problem. Detect unusual events in live video **correctly** and **in real time**, using a **fixed amount of compute**, and explain each alert in a way that **can be checked**.

Why it is hard

24 GB

of video a day, from one camera

→ too much to watch

< 8%

of UCF-Crime segments are abnormal



→ rare, imbalanced

a score

is all a detector returns

→ no what, no why

Sources: video volume – Xu et al., EuroSys 2019; UCF-Crime segment counts – Acharya et al., CVPR 2026.

Related work



Line of work	What it does	Examples
Weakly supervised VAD	Knows which video has an anomaly, not when	Sultani et al. (2018), REWARD (2024), VadCLIP (2024)
Training-free VAD	Uses pretrained models as they are, no training	LAVAD (2024), MoniTor (2025), PANDA (2025)
Explainable VAD	Describes the event in natural language	VAD-LLaMA (2024), AnomalyRuler (2024), Holmes-VAU (2025)
Fast/slow cascades	Cheap model first; large model only when needed	SlowFastVAD (2025), Flashback (2025), AssistPDA (2025)

ARGUS **uses** a fast/slow cascade too. What we change is how the cascade is controlled.

Limitations of existing systems



Compute use is not controlled

A threshold decides **which** segments go to the large model, **not how many**.

Alerts are slowest when it matters

If the alert waits for the large model, it comes **late during an event**.

Explanations cannot be checked

A model can write a confident sentence that is **wrong**, and **nothing checks it**.

Every camera is treated the same

A forklift is normal at a loading dock but **alarming** in a corridor at night.

Our idea: treat the cascade as a **scheduler**, not a switch.

A switch says yes or no to each segment. A scheduler decides how many, and when.

Motivation

VLMs can explain

Vision-language models describe *what* happened in the frames, and *why*.

LAVAD, VAD-LLaMA,
Holmes-VAU



But they are slow

These models are too slow to look at 25 frames every second on a low-cost device next to the camera.



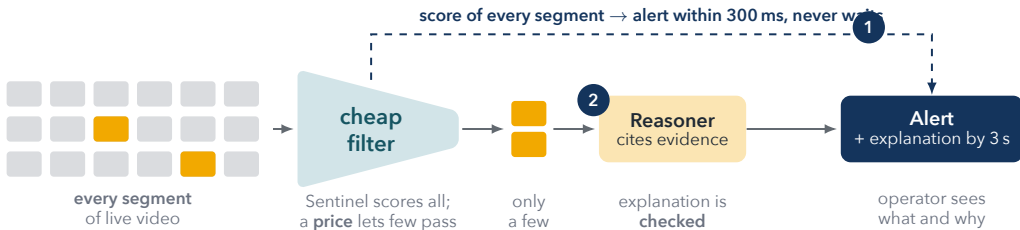
So: a cascade

A cheap detector watches everything and calls the large model only when a segment looks suspicious.

SlowFastVAD, Flashback,
MoniTor

Objectives

Build and test ARGUS: **fast alerts** and **explanations that can be checked**, on low-cost hardware.



1 The alert never waits

Alert probability within a **hard deadline**; the explanation comes **later**.

2 Explanations that can be checked

A **frozen** model must **cite evidence**; every citation is **checked**.

Problem formulation

A policy ϕ decides, per segment, whether to escalate and what probability to report.

$$\min_{\phi \in \Phi} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \ell(\hat{p}_t^\phi, y_t) \quad \text{s.t.} \quad \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T a_t^\phi \leq B, \quad \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{I}[L_t^\phi > \Delta_{\text{hard}}] \leq \epsilon$$

Accuracy

Brier score $(p - y)^2$: a proper scoring rule, so **calibration** counts, not only ranking.

Compute

At most a fraction B reaches the expensive model, B **fixed by the hardware**.

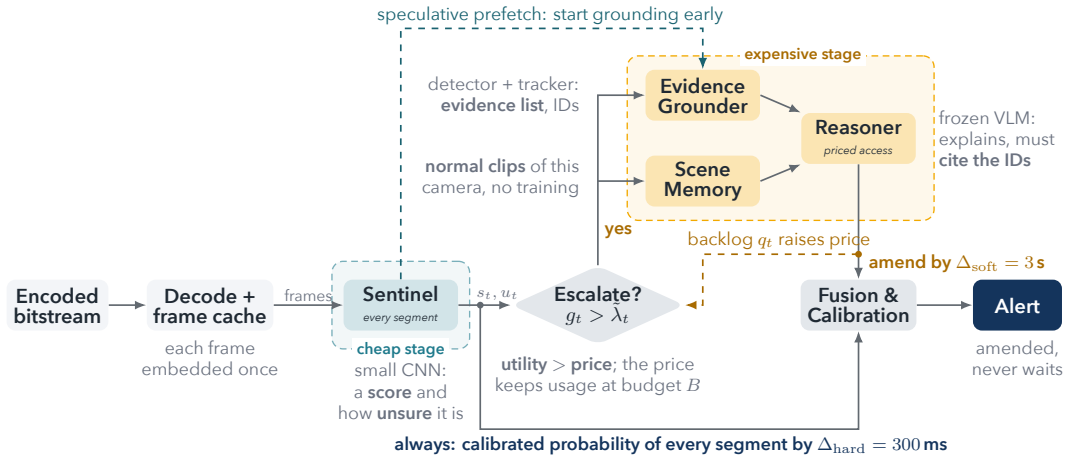
→ Escalation controller

Latency

At most $\epsilon = 1\%$ of segments miss the **300 ms** deadline.

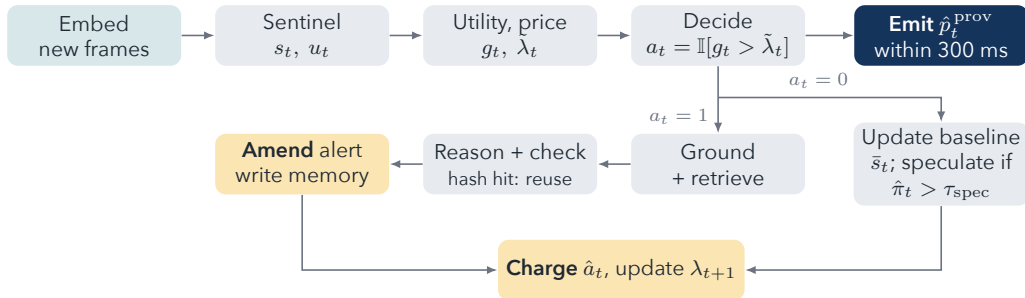
→ Fusion layer

System overview



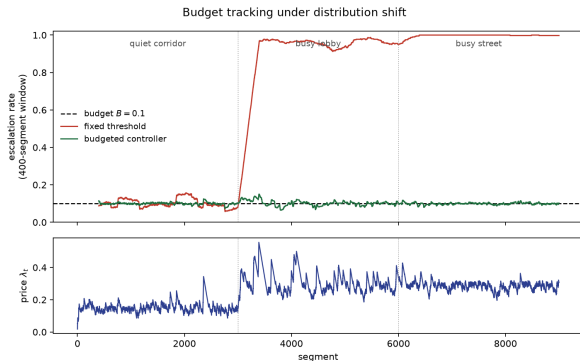
The complete procedure – one segment

ALWAYS, BEFORE ANYTHING CAN WAIT



The score leaves **before** any branch that could block. The baseline moves only when $a_t = 0$, so it never drifts toward an event.

Simulation results – the price holds the budget



Escalation rate

Budget 10%; threshold tuned on the first scene only

Scene	Threshold	Price
quiet	10.0%	10.2%
lobby	96.0%	10.3%
street	99.9%	9.9%
worst gap	89.9 pp	0.3 pp

Simulation: generated scores, not video – tests the controller, **not the detector**. **Next:** real video.

Summary

1 Problem

Catch unusual events in live video **fast**, on a **fixed compute budget**, with explanations that **can be checked**.

2 Gap

A **threshold** cannot control **how many** segments reach the large model; alerts **wait**; explanations go **unchecked**.

3 ARGUS

The **Sentinel** scores every segment, a **price** picks the few for the **Reasoner**, which must **cite evidence**, and the alert **never waits**.

4 First result

In simulation, the price holds usage at **about 10%** on every scene; a threshold jumps to **96%**.

ARGUS turns the cascade from a **switch** into a **scheduler**.

References – video anomaly detection

- [1] Sultani, Chen, Shah. *Real-world anomaly detection in surveillance videos*. CVPR 2018
- [2] Karim, Doshi, Yilmaz. *Real-time weakly supervised video anomaly detection (REWARD)*. WACV 2024
- [3] Wu et al. *VadCLIP: adapting vision-language models for weakly supervised VAD*. AAAI 2024
- [4] Zanella et al. *Harnessing large language models for training-free VAD (LAVAD)*. CVPR 2024
- [5] Yang et al. *MoniTor: LLMs with instruction for online VAD*. NeurIPS 2025
- [6] Yang, Gao, Shou. *PANDA: towards generalist VAD via agentic AI engineer*. NeurIPS 2025
- [7] Lv, Sun. *Video anomaly detection and explanation via LLMs (VAD-LLaMA)*. arXiv 2024
- [8] Yang et al. *Follow the rules: reasoning for VAD with LLMs (AnomalyRuler)*. ECCV 2024
- [9] Zhang et al. *Holmes-VAU: long-term video anomaly understanding*. CVPR 2025
- [10] Ding et al. *SlowFastVAD: simple detector and RAG-enhanced VLM*. arXiv 2025
- [11] Lee et al. *Flashback: memory-driven zero-shot, real-time VAD*. arXiv 2025
- [12] Yang, Gao, Shou. *AssistPDA: an online video surveillance assistant*. arXiv 2025

References – methods and data

- [13] Sensoy, Kaplan, Kandemir. *Evidential deep learning to quantify classification uncertainty*. NeurIPS 2018
- [14] Page. *Continuous inspection schemes (CUSUM)*. Biometrika 1954
- [15] Neely. *Stochastic network optimization with application to communication and queueing systems*. Morgan & Claypool 2010
- [16] Brier. *Verification of forecasts expressed in terms of probability*. Monthly Weather Review 1950
- [17] Kwon et al. *Efficient memory management for LLM serving with PagedAttention*. SOSP 2023
- [18] Leviathan, Kalman, Matias. *Fast inference from transformers via speculative decoding*. ICML 2023
- [19] Xu, Botelho, Lin. *VStore: a data store for analytics on large videos*. EuroSys 2019
- [20] Acharya et al. *The road less seen: segment exploration for weakly supervised VAD*. CVPR 2026

Thank you



Questions?

ARGUS • Md. Shahriar Kabir • Tawyabul Islam Tamim • Supervisor: Md. Ahasanul Alam